DRUMPANEL · RISK REGISTER AND HAZARD ANALYSIS

DrumPanel risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First risk register, preliminary hazard analysis and first-pass design FMEA (kit 1.8 upgrade of a TRL 3 repo, stage P3), for option A, the current bench set

DrumPanel risk register and hazard analysis

The top hazard is an explosion when a drum that still holds flammable vapour is cut; it is controlled by the acceptance rule, the water purge and a pumped LEL reading below 5 % at both bungs on the day of cutting, and it stays the first safety stop of the build plan. The next hazards are fingers in the in-running nips of the discs, rolls and chains (controlled by guards with 8 mm slots that need a tool to remove), cut edges, and a drum full of water (never lifted). The top project risk is that no co-design partner is found, so nothing is built or measured; the highest design FMEA scores are a drum passed by a detector that was not bump-tested and a bending roll set to the wrong dial reading. Method: STANDARDS section 23 and ISO 12100 (as guidance for a non-certified hand tool). Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (sure to be found) to 5 (unlikely to be found).

> **Safety:** DrumPanel works on used oil drums, which can hold flammable vapour until they are purged and checked, and its hand-cranked tools have in-running nips at the slitting discs, the end cutter, the slip roll and three chain drives. Cut drum steel has sharp edges. A drum full of water weighs about 240 kg. Nothing in this register replaces a hazard review of the built prototype at TRL 4.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No co-design partner	Candidates named (a fair trade partner of Haitian metal artists, a vocational welding school), none approached; security in Croix-des-Bouquets	URS and PRS stay desk drafts; no TRL 4	3	5	Amish	Approach when TRL 4 opens; the design does not depend on one site
P2	Throughput short of PR-007	Task times are estimates; ring passes and handling dominate	About 2.20 drums an hour against 2.5	3	3	Design lead	Timed half-day trial decides (DMP-DDR-004); rail stops and batch notching as ideas (DMP-DEC-001)
P3	Chime notch force higher than estimated	The five-layer chime seam varies; 24.3 kN is a screening estimate	Lever force above 150 N	3	3	Design lead	Scrap-chime test before the punch dies are hardened (DMP-DEC-001, item 8); a longer ratchet lever as fallback
P4	Detector support not available locally	Calibration gas and service need a specialist supplier	Bump tests skipped; the safety case weakens	3	5	Partner	One detector shared by a cooperative with a service plan; no cutting without a bump test that day (safety stop)
P5	Drum supply mixed or unlabelled	Drums come from many sources	Many drums rejected; pressure to accept unknown drums	3	4	Partner	Acceptance rule held without exception (PR-013); record of drums turned away
P6	Cost over the value-engineering target	Detector, discs and punch dies dominate	USD 3,566 against USD 3,000	4	2	Amish	A hypothetical target; savings listed in DMP-DEC-001 (shared detector, stock slitter knives, ground shafting)
P7	No lathe with 1.1 m between centres nearby	Rolls are 1,092 mm long	Rolls bought in or made away	2	2	Partner	Buy ground shafting and turn only the journals (DMP-DEC-001)

Preliminary hazard analysis

Table 2. Hazards (residual after the controls in the design).

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Explosion or flash fire in a drum	Residual vapour from a drum that held fuel, solvent or an unknown liquid, or an oil drum not purged; friction at the cut	Burns, blast injury, death	5	3	Acceptance by label (PR-013; fire-resistant hydraulic fluids refused); water purge; LEL channel calibrated for heavy vapour (fix F3); pumped LEL reading below 5 % at both bungs on the day of cutting (PR-002); no grinder, torch or welder near a drum; the cut stays far below autoignition (DMP-STM-001 S17)	2
H2	Fingers drawn into the slip roll nip	Feeding panels by hand	Crushed fingers	4	3	Nip guards with 8 mm slots 45 mm from the nip (PR-010); guards need a tool to remove; one person on the crank, nobody else at the in-feed	2
H3	Fingers at the slitting discs or the end cutter wheel	Steadying the drum or the cut edge while cranking	Cuts, crushing	4	3	Disc guard with a front skirt 8 mm over the sheet; wheel cover; crank 155 mm and 150 mm from the cut; one person cranks, the other keeps hands on the drum away from the head	2
H4	Fingers in a chain drive	Reaching into the drive while cranking	Crushed fingers	4	2	Shear head chains inside the closed drive case; slip roll chain and gears in guard boxes held by screws (DMP-DDR-006, Q10)	1
H5	Cut edges	Handling panels, rings, hoop strips and head discs	Lacerations	3	4	Cut-resistant gloves EN 388 level C for every task with cut steel (BOM line 24); deburr every edge before stacking	2
H6	Back injury	Lifting the 33.7 kg cradle frame, the shear head or ring head, or a drum; tipping a water-filled drum	Back strain	3	3	Two-person rule over 25 kg, and for lifting either head to the rail (PR-011); a full drum is filled and drained in place, never lifted	2
H7	Repetitive strain	Cranking and levering for a working day	Shoulder and wrist strain	3	3	Forces at or below 150 N (PR-006); two people swap stations; an EN 1005-3 assessment at TRL 4	2
H8	Skin contact with oil residue and detergent	Draining and purging	Dermatitis	2	3	Nitrile gloves and aprons (BOM line 24); detergent biodegradable	1
H9				3	3		2

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
	Purge water or oil on the ground	Spills, overflowing tray, settling drum not emptied	Soil and water pollution			60 L tray under the bung; settling drum with a tap; skimmed oil to a recycler (PR-014); disposal route to agree with the partner (DMP-REG-001 G11)	
H10	Noise	Setting panel ends with a mallet	Hearing damage over years	2	2	Dead-blow mallet; ear defenders (BOM line 24)	1
H11	Drum rolls off the cradle	Side push while slitting	Crushed feet, falling drum	3	2	Screw brake on for the slit; a side push of 122 N rolls an empty drum, so the brake is always on for the slit (DMP-CAL-001 section 6)	1
H12	Ring head or shear head drops while being bolted on	Heavy head held at arm's length	Struck hands or feet	3	2	Second person steadies it; spacer plate bolts tight before cranking (build plan safety stop S7)	1
H13	Welding fume and burns while building the set	Building the frames	Fume, burns, eye injury	3	2	No drum in the workshop while welding or grinding; ventilation; welding PPE (build plan safety stop S1)	1

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Vapour check kit	Flag a drum with vapour before cutting	Reads low	A drum with vapour is passed (H1)	Detector not bump-tested; sensor poisoned; probe not deep enough in the bung; a methane-calibrated catalytic sensor under-reading heavy vapour	5	2	4	40	LEL channel calibrated on pentane or corrected for heavy hydrocarbons (STEM fix F3); bump test every day of use, logged; 1 m probe to the bottom of each bung; read both bungs; repeat if a drum stands overnight
F2	Bending roll setting	Leave panels flat within 10 mm over 1 m	Set to the wrong dial reading	Panels bowed; rework	The right reading changes from drum to drum (STEM fix F1); dials slipped; screws not locked	2	3	2	12	Check the first panel of every drum with a straightedge and trim the dial; pass a bowed panel again, turned over if over-bent; lock nuts
F3	Slitting discs	Shear the wall cleanly	Chipped or blunt edge	Torn cut, high crank force	Hitting a hoop ridge or chime not notched; wear	3	3	2	18	Notch all four points before slitting (build plan); inspect the discs daily; regrind or replace
F4	Driven disc traction	Pull the head along the cut	Slips on a thick drum	Slow slit; operator pushes the trolley	1.5 mm wall; traction margin 1.17 (DMP-STM-001 S13)	2	3	1	6	Push the trolley by hand while cranking; confirm the margin at TRL 4 (DMP-DEC-001 item 7)
F5	Lever notching punch	Cut the chime notch and hoop slot	Punch does not get through the chime	Notch incomplete; lever force rises	Chime thicker or harder than assumed	3	3	1	9	Scrap-chime test before hardening (DMP-DEC-001 item 8); longer lever as fallback
F6	Pivot pin and cross bolt (DMP-DDR-006, Q1)	Hang the shear head from the trolley	Cross bolt loosens	Head drops onto the drum (H12)	Vibration of the crank; nut not tight	4	2	2	16	Nyloc or double nut; check the bolt with the index pin before each shift
F7	Guards	Keep fingers from nips	Guard left off	Fingers reach a nip (H2 to H4)	Removed for cleaning and not refitted	4	2	2	16	Screws need a tool; the crank stays off until the guard is back (safety stop); daily check list

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F8	Drain stand chocks	Hold a full drum sloping to the bung	Chock splits	Drum shifts on the stand	Wet timber, coach screw loose	3	2	2	12	Hardwood, oiled; screws checked; drum filled and drained in place only
F9	Slip roll pinch gears	Keep the pinch rolls turning together	Teeth jump	Panel skews	Pinch opened beyond 61.5 mm centres	2	2	2	8	Set the pinch with a 1 mm sheet strip; gear centres checked when bought (DMP-DEC-001 item 5)
F10	Shear head chains	Drive the upper disc from the raised crank	Chain jumps or breaks	Crank spins free; slit stops	Slack chain; wear	2	2	3	12	3 mm slack at assembly; inspect through the case at each service
F11	Coupling to the ring head	Drive both upper discs together	Dog disengages	Ring head disc idles; one cut of the pair incomplete	Spacer plate bolts loose; 233 or 234 mm slot set wrong	2	2	2	8	All four spacer bolts tight before cranking (safety stop S7); both discs seen turning
F12	Drum brake	Hold the drum still for the slit	Pad slips	Drum turns; slit wanders (PR-004)	Worn pad; oil on the drum	2	2	2	8	Wipe the pad seat; replace the pad when hard